

Enhancing Bankruptcy Prediction with Automated Learning and Interpretability: A Comparative Study of Metaheuristics, XGBoost, and AutoML

An In-Depth Analysis of Metaheuristic Models for Bankruptcy Prediction

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This study explores a multi-stage approach to bankruptcy prediction by combining traditional metaheuristic methods with modern Automated Machine Learning (AutoML) and interpretable ensemble models. Initially, an artificial neural network (ANN) trained with a hybrid Particle Swarm Optimization (PSO) and Magnetic Optimization Algorithm (MOA) showed poor recall despite high accuracy, highlighting the challenges of imbalanced datasets. These initial experiments were conducted using a Taiwanese financial dataset. Subsequent experiments focused on resampling techniques such as Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN), and Generative Adversarial Networks (GANs) to improve minority class prediction. A Random Forest model trained on the resampled data significantly improved recall. To further advance performance and interpretability, we implemented eXtreme Gradient Boosting (XGBoost) and AutoGluon on the Polish Bankruptcy dataset. XGBoost achieved strong recall and interpretability via SHapley Additive exPlanations (SHAP) values, while AutoGluon leveraged ensemble-based AutoML with permutation importance to deliver near acceptable performance. Comparative analysis shows that XGBoost provides better transparency, whereas AutoGluon offers automation with minimal tuning effort. This paper presents a comprehensive evaluation of each approach, providing actionable insights for practical financial risk modeling.

CCS CONCEPTS • Machine Learning • Data visualization • Machine learning • Neural networks • Ensemble Model • Imbalanced learning • Metaheuristics • Model Interpretability

Additional Keywords and Phrases: Bankruptcy Prediction, ANN, Random Forest, PSO, MOA, GAN, SMOTE, ADASYN, XGBoost, AutoGluon, SHAP, Permutation Importance

1 INTRODUCTION

Corporate bankruptcy continues to pose a major concern in financial risk management, as its damage can create a ripple effect on supply chains, credit markets, and entire economies, while also disrupting the operations and livelihoods of a single firm's stakeholders. Better bankruptcy prediction technologies and techniques can increase the general stability of financial systems, enable prompt interventions, and protect impacted individuals. Despite being foundational, traditional statistical methods, such as Beaver's univariate ratios (1966), Altman's Z-score (1968), logistic regression and discriminant analysis have frequently been limited in their ability to handle complex, nonlinear data distributions due to assumptions like linearity and normality [1].

To address these limitations, this paper investigates advanced machine learning strategies for bankruptcy prediction. The first phase of the study involves experiments conducted on a Taiwanese financial dataset, where artificial neural networks (ANNs) were trained using metaheuristic algorithms, namely Particle Swarm Optimization (PSO) and the Magnetic Optimization Algorithm (MOA). These models serve as a foundation for exploring the impact of optimization strategies in the absence of gradient descent. However, due to their inability to handle imbalanced class distributions effectively, we expanded our investigation to ensemble-based models.

The study then shifts focus to Random Forest classifiers augmented with resampling techniques such as Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN), and Generative Adversarial Network (GAN)-based synthetic data generation. These methods address the prevalent issue of class imbalance and significantly improve model recall—an essential metric in high-risk domains. We further apply our methodology to the Polish Bankruptcy dataset and introduce eXtreme Gradient Boosting (XGBoost) and AutoGluon as more advanced alternatives. XGBoost enables fine-grained feature interpretation via SHapley Additive exPlanations (SHAP) values, while AutoGluon offers a high-performance Automated Machine Learning (AutoML) pipeline with minimal user intervention. A comparative evaluation of both models highlights trade-offs between automation and interpretability, concluding with practical recommendations for deployment in financial analytics environments.

2 PROBLEM FORMULATION

This research paper addresses the question of how machine learning models can be optimized to predict corporate bankruptcy accurately while supporting timely, data-driven business decisions. As previously mentioned, from a business perspective, the inability to accurately anticipate a company’s bankruptcy poses a risk not just to those within the company, but all its stakeholders (especially investors). The integration of machine learning into existing models offers a promising solution that is not without its caveats. While this integration may enable businesses to leverage financial data and identify the patterns and trends that lead to bankruptcy, it must provide comprehensive insights that are actionable for the decision-makers of a firm. This means that data obtained from high-predictive models must be interpretable, of high quality, and applicable to real-world financial scenarios.

3 IMPLEMENTATION WITH TAIWANESE DATASET

This section provides an overview of the dataset used in our study, along with the modeling approaches employed. We begin by describing the structure and characteristics of the dataset, followed by implementations of an ANN enhanced with metaheuristic optimizers and a Random Forest model utilizing resampling techniques. Subsequent sections detail the hyperparameter tuning strategies used and provide a comparative analysis of model performance based on key evaluation metrics.

3.1 Dataset Overview

3.1.1 Source and Context

The dataset used in this study comprises firm-level financial data originally curated for bankruptcy prediction research. It contains 6,819 corporate entries collected across different fiscal periods and includes both bankrupt and non-bankrupt Taiwanese companies [2]. This dataset is publicly used in financial distress prediction research and serves as a solid benchmark for machine learning classification tasks.

3.1.2 Target Variable and Feature Categories

The dataset consists of 95 numerical financial indicators and 1 binary target variable labeled "Bankrupt?", with 1 referring to bankrupt, 0 referring to non-bankrupt. The input features span a range of financial dimensions, including profitability metrics (return on assets and net income ratios), liquidity ratios (current and quick ratios), leverage and solvency indicators (debt ratio and equity-to-liability), and efficiency metrics (asset and inventory turnover rates). Additional categories include cash flow metrics (cash flow to sales), growth and market indicators (revenue growth, earnings per share), and composite financial flags.

3.1.3 Class Imbalance and Preprocessing

A notable challenge with this dataset is its high class imbalance, since only around 6% of the companies labeled bankrupt are present. This disparity makes it essential to apply resampling strategies or use evaluation metrics that can better handle skewed distributions, such as the F1-score or AUC.

Fortunately, the dataset was clean and fully numeric, with no missing values, allowing for its immediate use in model training without the need for any imputation or categorical encoding. The only additional preprocessing step taken was the scaling of all training features.

3.2 ANN with PSO and MOA

3.2.1 Model Overview

The first model explored in this study was an ANN, a supervised learning algorithm inspired by the structure of the human brain. ANNs are composed of interconnected layers of nodes (neurons) that transform inputs through weighted connections and nonlinear activations to perform tasks such as classification. This implementation was guided by prior work demonstrating that training ANNs using hybrid metaheuristic algorithms—specifically the MOA combined with PSO algorithms—can significantly improve both prediction accuracy and training efficiency for bankruptcy prediction tasks [1].

The architecture implemented consisted of an input layer, a single hidden layer with 10 neurons, and an output layer with one neuron for binary classification. The forward pass of the network applies the hyperbolic tangent (tanh) activation function to the hidden layer and a sigmoid activation function to the output layer to produce probability-like outputs suitable for binary decisions. This structure was instantiated using the class: `ANN(input_size, hidden_size=10, output_size=1)`

During the forward pass, the network applies a tanh activation function to the hidden layer and a sigmoid activation function to the output, which is standard for binary classification tasks. Instead of traditional gradient-based backpropagation, the model was trained using a hybrid metaheuristic optimization approach combining elements of PSO and MOA. These algorithms search the weight space by simulating swarm behavior and magnetic attraction rather than relying on gradient descent.

Despite its conceptual novelty, the ANN performed poorly in practice. While it achieved high accuracy due to the imbalanced nature of the dataset, its recall was extremely low, indicating that it frequently failed to identify actual bankrupt cases. This confirmed that the model was not learning meaningful decision boundaries and instead leaned heavily toward predicting the majority class. As a result, the ANN served as a useful baseline but was ultimately replaced by more effective models, including Random Forest with resampling, which significantly improved minority class detection.

3.2.2 Hybrid PSO-MOA Optimization

To train the ANN model, a hybrid metaheuristic approach was employed, combining PSO and MOA. Unlike traditional training methods that rely on backpropagation and gradient descent, this technique evolves network weights by optimizing a fitness function over multiple iterations using a swarm-based strategy.

PSO is inspired by the social behavior of flocks of birds or schools of fish, where each candidate solution (or particle) adjusts its position in the search space based on its own best-known position and that of the swarm. MOA complements this with the concept of magnetic attraction, pulling solutions toward a global best using a force-based update mechanism.

In the implementation (`hybrid_moa_pso(...)`), each particle represents a flattened vector of ANN weights. The swarm initializes randomly, and the fitness of each particle is evaluated using mean squared error (MSE) between the model's predicted probabilities and the true bankruptcy labels. This is handled by the `mse_loss()` function. During optimization, the velocity of each particle is updated through a combination of three components: (1) cognitive influence from its own best position, (2) social influence from the global best solution, and (3) magnetic force, a simplified MOA-inspired vector that nudges particles toward the global best.

The weights are then passed through the ANN's `forward()` method, which applies a tanh activation in the hidden layer and a sigmoid function in the output layer to produce probabilistic predictions for binary classification. The optimization continues for a predefined number of iterations, with particles updating their positions based on the combined PSO-MOA dynamics.

This hybrid strategy provided a gradient-free alternative to backpropagation, aimed at finding an optimal set of weights without relying on differentiable loss landscapes. However, despite the novelty of this training method, the ANN's recall remained extremely low due to the class imbalance in the original dataset, ultimately leading to the adoption of resampling strategies and more suitable models like Random Forest.

3.2.3 Data Preprocessing

The bankruptcy dataset was first loaded and preprocessed to ensure consistency across all model training pipelines. The target label `Bankrupt?` was separated from the feature set, with the remaining columns constituting the input matrix `X`. To bring all features onto a uniform scale, `MinMaxScaler` was applied, normalizing values to the `[0, 1]` range. This scaling step was crucial for models sensitive to feature magnitude, such as neural networks and distance-based algorithms. Importantly, feature scaling was performed before splitting the data, maintaining consistency across all experiments and ensuring that the train-test split did not leak information. This same preprocessing approach was reused consistently for all resampling strategies (`ADASYN`, `SMOTE`, and `GAN`) and model evaluations to ensure comparability across different experimental conditions.

3.2.4 Results and Limitations

For model evaluation, standard classification metrics such as accuracy, precision, recall and F1 score were all evaluated, along with a confusion matrix, allowing us to observe not only overall correctness (accuracy) but also the model's ability to detect the minority class, which in this case represents bankrupt firms.

Despite achieving a high accuracy of $\sim 96\%$, the model's performance was critically poor in identifying bankrupt companies, as seen in the table below:

Table 1: ANN Performance with PSO and MOA Optimization Algorithms

Precision	Recall	F1-Score	Accuracy	Confusion Matrix
0.333	0.020	0.037	0.962	$\begin{bmatrix} 1311 & 2 \\ 50 & 1 \end{bmatrix}$

The confusion matrix clearly shows that while the model correctly classified 1,311 non-bankrupt firms, it missed 50 out of 51 bankrupt firms, resulting in a recall of just 1.96%. This is a severe failure, especially in a financial context where detecting bankruptcies is the primary objective.

This misleadingly high accuracy was a direct result of class imbalance, where the majority class (non-bankrupt) dominated the dataset. As a result, the ANN largely defaulted to predicting the majority class, classifying nearly all samples as non-bankrupt. A more robust solution requires resampling strategies, such as SMOTE, ADASYN, or GAN-based augmentation, to address the class imbalance issue, which we will explore in subsequent experiments.

3.2.5 Key Takeaways and Insights

The hybrid metaheuristic-trained ANN (MOA + PSO) was an informative experiment, but ultimately ineffective for this imbalanced binary classification task. Despite achieving high accuracy, the low recall value indicates that it failed to identify bankrupt firms, highlighting the inadequacy of accuracy as a metric in this context. Early training issues, such as sigmoid overflows and unstable weight updates, also suggested deeper architectural or initialization problems.

These challenges motivated a shift toward Random Forest models paired with resampling techniques like SMOTE, ADASYN, and GAN-based methods, which yielded far better recall and F1 scores. This experience reinforced the importance of prioritizing recall in high-stakes domains like bankruptcy prediction.

3.3 Random Forest

3.3.1 Model Overview

Random Forest is a supervised ensemble learning algorithm that combines the predictions of multiple decision trees to improve classification accuracy and generalization. It operates by training each tree on a random subset of the data and features, then aggregating their outputs through majority voting. This approach reduces variance and helps prevent overfitting, making Random Forest particularly robust for complex, real-world datasets.

The bankruptcy classification task involves identifying rare events, as the dataset is highly imbalanced with a small proportion of bankrupt companies. Random Forest was chosen not only for its robustness and interpretability but also because it performs well on imbalanced data, especially when combined with resampling techniques such as SMOTE, ADASYN, or synthetic data generation.

The model was implemented using the `RandomForestClassifier`, with the following baseline configuration: `RandomForestClassifier(n_estimators=100, random_state=42)`

This setup specifies the use of 100 trees for a strong ensemble effect and fixes the random seed to ensure reproducible results.

Preliminary comparisons with the ANN showed that Random Forest achieved better recall, which is an important metric in this context, as it reflects the model's ability to correctly identify actual bankruptcies. This advantage, along with its flexibility and ease of tuning, reinforced the decision to use Random Forest as a core model in this study.

3.3.2 Experimental Setup

To ensure consistent and reliable evaluation of the Random Forest classifier, a structured experimental pipeline was followed. This involved a clear sequence of preprocessing steps, model validation using cross-validation, and a comprehensive set of evaluation metrics to assess model performance.

All feature data was standardized using feature scaling (MinMaxScaler) before performing the train-test split. Standardization ensures that each feature contributes equally during training and prevents variables with larger numeric ranges from dominating the learning process. This step was uniformly applied to all datasets used in the experiments to maintain consistency across model inputs.

Model evaluation was conducted using 5-fold cross-validation, implemented with the KFold function from scikit-learn. This strategy reduces the likelihood of overfitting and provides a more robust and generalized estimate of model performance across different subsets of data.

The evaluation relied on five key metrics: Receiver Operating Characteristic Area Under the Curve (ROC AUC) (one-vs-rest), macro-averaged precision, recall, F1-score, and accuracy. ROC AUC evaluates the model's ability to distinguish between classes, which is especially important for imbalanced datasets. Precision measures the proportion of correct positive predictions, while recall reflects the ability to capture all actual positives, which is crucial in high-risk domains like bankruptcy prediction. F1-score combines precision and recall into a single balanced metric. Although accuracy measures overall correctness, it can be misleading on skewed datasets, so it was considered alongside the other metrics.

These metrics were specified using a scoring dictionary and passed into scikit-learn's `cross_validate` function. This function performed training and evaluation across all five folds, computing each metric separately for every split. The results from each fold were then averaged, producing a final set of performance scores that offer a reliable and stable summary of how well the model is expected to perform on unseen data.

3.3.3 Baseline Model without Resampling

The baseline Random Forest model was first trained and evaluated using the original, imbalanced dataset without applying any resampling techniques.

Despite achieving a high overall accuracy of approximately 0.97, the baseline model's recall was notably low, around 0.545. This means that the model failed to correctly identify a substantial portion of actual bankrupt cases, which is a critical shortcoming in the context of bankruptcy prediction.

This performance gap highlighted the need to address the class imbalance issue. The inability of the baseline model to adequately detect bankrupt firms underscored the importance of incorporating resampling methods to balance the training data. By improving the representation of the minority class, these techniques aim to enhance the model's ability to generalize and capture rare but important instances.

3.3.4 Resampling Techniques Applied

Resampling is essential in imbalanced classification problems, as models trained on skewed data tend to favor the majority class, leading to poor recall and low sensitivity to rare but critical outcomes such as predicting bankruptcy. Three resampling methods were implemented: ADASYN, SMOTE, and a GAN-based approach.

ADASYN works by generating synthetic data points specifically for harder-to-learn minority class instances. It adaptively focuses on samples that are sparsely represented or lie near decision boundaries, helping the model generalize better in these challenging areas [3].

SMOTE creates new synthetic examples by interpolating between existing minority class samples based on feature-space similarities. Unlike ADASYN, SMOTE treats all minority samples equally without emphasizing more complex cases [4].

For both ADASYN and SMOTE, the `fit_resample()` method was applied after scaling the features to ensure consistent input ranges. This step ensured the resampling process respected the underlying feature distributions.

The GAN-based resampling involved a more advanced technique using adversarial learning. The implementation included a function `getMinorityClass()` to isolate the minority class, followed by constructing two neural networks: a generator that produces synthetic data and a discriminator that learns to distinguish between real and generated samples. The two networks were trained in tandem using cross-entropy loss, allowing the generator to improve over time in producing realistic samples. The GAN model was pre-coded and later fine-tuned to align with the structure of the bankruptcy dataset [5].

These resampling techniques were applied before training the Random Forest model, significantly enhancing its exposure to minority class patterns and laying the groundwork for improved recall and balanced performance.

3.3.5 Resampled Model Performance

With resampling techniques applied, the Random Forest model showed significant improvement in its ability to detect bankrupt companies, particularly in terms of recall. Identifying as many actual bankrupt firms as possible is far more important than optimizing precision or overall accuracy, as missing a potential bankruptcy poses greater risk than issuing a false alarm.

The recall for the baseline model was approximately 0.545, indicating poor sensitivity to the minority class. After resampling, all three methods produced substantial improvements:

Table 2: Random Forest Performance with and without Resampling Techniques

Method	ROC AUC	Precision	Recall	F1-Score	Accuracy	Training Time (s)
Baseline	0.880	0.720	0.545	0.615	0.930	18.090
ADASYN	0.912	0.821	0.837	0.740	0.926	50.619
SMOTE	0.905	0.785	0.655	0.714	0.934	49.805
GAN	0.924	0.905	0.885	0.787	0.939	16.057

While slight trade-offs were observed in metrics like precision and accuracy compared to the baseline, these are justifiable. In bankruptcy prediction, recall is prioritized since it is more acceptable to flag a financially stable company than to overlook one that is at risk.

Notably, the GAN-based model not only achieved the highest recall (~0.885) but also had the shortest training time (16 seconds) among all resampled models, despite its more complex training mechanism. This suggests that, once properly fine-tuned, GAN-generated data can be both realistic and computationally efficient, which is an important consideration for scalable or real-time deployment scenarios.

Overall, resampling proved essential to achieving balanced performance and making the classifier suitable for high-stakes decision-making in financial risk prediction.

3.3.6 Key Takeaways and Insights

This section highlights the transformative impact of resampling techniques on the reliability of Random Forest models in detecting minority class instances, specifically bankrupt companies. Without resampling, the baseline model performed well in terms of accuracy but failed to capture a significant portion of actual bankruptcies. Once the dataset was rebalanced using ADASYN, SMOTE, and GAN-based augmentation, recall improved dramatically across all methods, making the model far more suitable for high-stakes financial risk prediction.

Among the resampling techniques, the GAN-based approach stood out by achieving the highest recall (~0.885) and the shortest training time (16 seconds). These results suggest that GANs, once fine-tuned, not only generate realistic synthetic samples but also do so efficiently, making them a strong candidate for deployment in real-world systems where both performance and speed are important. As long as interpretability is not the primary concern, GAN-based resampling is recommended for scenarios where recall is critical.

That said, the current approach is not without limitations. The Random Forest model, while effective, does not inherently address class imbalance without external resampling. Additionally, the trade-off between interpretability and complexity becomes relevant when using synthetic data, particularly from GANs. Understanding which features drive predictions becomes more difficult, which may be a concern in regulatory or audit-focused financial environments.

Future work will explore combining resampling with feature selection or cost-sensitive learning, aiming to improve performance while reducing noise and computation. Another planned enhancement involves incorporating SHAP to analyze feature importance and better understand which financial indicators most strongly influence bankruptcy predictions. This could help bridge the gap between predictive power and interpretability, making the system not just accurate, but actionable.

3.4 Hyperparameter tuning

Given the ANN and RF baseline machine learning models, hyperparameter tuning was done for further improvement. Hyperparameters control the learning process of the model; tuning them properly would lead to better predictive results. Two methods are done to achieve this: Grid Search (GS) and Differential Evolution (DE).

GS takes lists of hyperparameters and evaluates all possible combinations of them. This method of tuning based on specified hyperparameter values is commonly used for its simplicity. However, GS fails to try hyperparameters outside of those given lists. Additionally, when the lists of hyperparameters are too big, GS becomes computationally expensive and inefficient as the numbers of configurations grow exponentially.

DE is a population-based metaheuristic approach to optimize finding the best solution in a high-dimensional continuous space. Given this, it is particularly useful when used for hyperparameter tuning, as it explores new hyperparameters as opposed to only trying hyperparameters in a specified list. However, since it is an optimization algorithm, it needs to converge into a specific set of values, which may require more iterations. DE is also sensitive to control parameters, unlike GS, that does not have any.

Given the different strengths and weaknesses of the two hyperparameter tuning methods, both were tested on the two machine learning models.

The tuning process can introduce overfitting to the model, especially when the model tries to capture every feature of the training data. Therefore, 5-fold cross-validation was used through all of the tuning trials. Cross-validation partitions

the data into multiple training and validation folds, ensuring that every data point is used for both training and validation. This could encourage the model to generalize better and reduce variance.

3.5 Results and Discussion

This section discusses the results achieved from each implementation and their hyperparameter tuned versions, taking the ANN + MOA + PSO hybrid model as the baseline for ANN, and the RF with resampling as the baseline for RF.

Table 3: Results of all Model Implementations for ANN and RF

Model Implementation	ROC AUC	Accuracy	Precision	Recall	F1 Score
Atiya 2001 ANN	0.004*	0.962*	0.333*	0.020*	0.037*
Baseline ANN	0.504	0.505	0.511	0.151	0.233
Baseline RF	0.880	0.930	0.720	0.545	0.596
ANN with DE	0.840	0.840	0.864	0.807	0.834
RF with DE	0.967	0.945	0.906	0.906	0.906
ANN with GS	0.858	0.858	0.845	0.874	0.860
RF with GS	0.925	0.939	0.787	0.571	0.606

*Estimated values from Atiya (2001), based on reported accuracy and dataset characteristics.

All other results represent mean values across 5-fold cross-validation with ADASYN resampling

The summary of bankruptcy prediction model performances includes a 2001 ANN model [6] for historical comparison; these metrics are taken by following its methodology combined with the same dataset. All other model performances were estimated using the mean 5-fold cross-validation on ADASYN-resampled datasets.

The baseline versions of ANN and RF already show improved results from the historical implementation with better metrics—including ROC AUC, precision, recall, and F1. A lower accuracy is expected due to the shift of focus to other metrics like F1 score and ROC AUC from the imbalanced nature of the dataset. To improve the two baseline models, hyperparameter tuning was used on both.

The DE-optimized ANN used `num_particles=112` and `max_iter=37` while its GS counterpart used `num_particles=150` and `max_iter=500`. Despite the marginal differences in metrics, GS ran much faster due to only having two parameter lists to exhaustively search. The RF-DE model stands out as the best-performing implementation overall, achieving a ROC AUC of 0.967 and top scores across precision, recall, and F1 metrics. This model used hyperparameters: `n_estimators` ≈ 198 , `max_depth` ≈ 4 , and `min_samples_split` ≈ 2

4 IMPLEMENTATION WITH POLISH DATASET

This section outlines the implementation pipeline for bankruptcy prediction using the Polish dataset, focusing on preprocessing, imbalance handling, and a comparative evaluation of XGBoost and AutoGluon models.

4.1 Dataset Overview

4.1.1 Source and Context

The dataset used in this section of the study is the “Polish Bankruptcy Dataset” [7]. It consists of financial statement data from Polish companies collected over a multi-year period spanning both bankrupt and non-bankrupt firms, with bankrupt entities analyzed from 2000 to 2012, and non-bankrupt entities tracked between 2007 and 2013. The dataset has a temporal structure, which means that there is a patterned organization of time [8]. The dataset is segmented into five subsets, each representing financial data from a specific year before the event of bankruptcy occurring. For example, the "1stYear" dataset contains features recorded five years before bankruptcy, while "5thYear" includes data from just one year prior. After merging, the final dataset consisted of 43,405 and 65 features.

4.1.2 Target Variable and Feature Categories

Each instance in the dataset includes a binary target variable labeled Bankrupt, which indicates whether a company eventually failed (1) or remained (0). Another column, YearsToBankruptcy, captures the forecast horizon representing how each row in the dataset shows what a company’s finances looked like 1 to 5 years before a possible bankruptcy happened — this helps the model learn early warning signs. However, despite the various features, the dataset is highly imbalanced with only about 6% of firms declaring bankruptcy.

Overall, the dataset is well-organized and includes both financial details and outcomes (bankruptcy or not), making it ideal for training models that can predict future bankruptcies and reveal which financial factors contribute most to a company’s collapse.

4.1.3 Class Imbalance and Preprocessing

To prepare for modelling, the five subsets of the Polish dataset were merged into one Dataframe. Since each dataset corresponded to a specific number of years before a bankruptcy event occurred, a new column named “YearsToBankruptcy” was added to indicate the prediction horizon. Prediction horizon denotes how far in advance the model attempts to predict an outcome.

Following the merge, the dataset contained several missing values across its 64 financial features. These were handled using imputation, a method that replaces missing entries with the average value of each respective feature.

One of the primary challenges with this dataset is the high-class imbalance, as only approximately 6% of the firms in the combined dataset are labeled bankrupt. To mitigate this issue, the dataset was split into training and testing sets using stratified sampling, a technique used to ensure that each subset (train/test) has the same proportion of classes as the original. An oversampling technique was then applied to the training set only, in order to enhance the model’s ability to learn from the minority class without introducing bias into the evaluation phase. The resampling method used was SMOTE. The method successfully rebalanced the class distribution in the training data, improving the models’ ability to detect rare bankruptcy cases in subsequent evaluations.

4.2 XGBoost

4.2.1 Model Overview

The XGBoost classifier was employed due to its robust performance on structured data and its ability to handle imbalanced classification through the `scale_pos_weight` parameter [9]. The model constructs decision trees sequentially, where each iteration corrects the errors of the previous one, enabling it to capture complex feature interactions. Its compatibility with SHAP further enhances its interpretability by quantifying the contribution of each feature to individual predictions.

4.2.2 Model Training and Optimization

To improve the model's ability to identify bankrupt firms, hyperparameter optimization was conducted using GridSearchCV, with recall as the primary scoring metric. The grid search explored combinations of `n_estimators`, `max_depth`, and `learning_rate`, ultimately selecting the configuration that yielded the best cross-validated recall on the SMOTE-resampled training data. The best model used 300 estimators, a learning rate of 0.2, and a maximum tree depth of 5. This tuned configuration was then used to train the final classifier for evaluation on the holdout set.

4.2.3 Evaluation

The final XGBoost model achieved a recall of 0.7177, a precision of 0.78, and an ROC-AUC of 0.9719 on the test set, indicating strong overall predictive performance. The confusion matrix confirmed that the model effectively distinguished between bankrupt and non-bankrupt firms, though a notable number of false negatives remained. To address this, threshold tuning was applied. At a lowered classification threshold of 0.01, the recall increased significantly to 0.9258—capturing nearly all bankruptcies at the cost of introducing more false positives. This flexibility makes the model suitable for high-risk financial screening, where recall is often prioritized over precision.

In terms of interpretability, SHAP values were used to assess feature contributions. The SHAP summary bar chart reveals that the most impactful features included profit on operating activities to financial expenses, operating expenses to total liabilities, and profit on sales to sales. Notably, these top features had substantially higher mean SHAP values than the rest, suggesting a concentrated dependency on a few key signals.



Figure 1: Top 10 Features Based on SHAP (XGBoost)

4.2.4 Results and Limitations

The model's predictions were further analyzed using SHAP to understand feature contributions. SHAP values revealed that ratios reflecting profitability, leverage, and operational efficiency had the most influence. Features such as "profit on operating activities to financial expenses" and "operating expenses to total liabilities" consistently appeared among the top contributors. SHAP's visualization capabilities also highlighted variance in model confidence, with wide SHAP distributions in low signal regions, suggesting that the model is less stable for borderline or ambiguous cases. Figure 1 illustrates the features that SHAP chose which contributed the most to the model.

4.2.5 Key Takeaways and Insights

The XGBoost model demonstrated strong predictive capabilities in identifying financially distressed firms, achieving high recall and AUC. Its performance improved significantly through threshold tuning, allowing practitioners to adjust the sensitivity of predictions based on business needs. A standout advantage of XGBoost was its integration with SHAP, which provided clear, interpretable explanations of model decisions.

Features such as profit on operating activities to financial expenses and operating expenses to total liabilities emerged as dominant risk indicators, reinforcing the model's transparency and utility for early warning systems, regulatory assessments, and credit risk auditing. However, optimizing recall did require additional post-training adjustments, and interpreting SHAP outputs meaningfully still demands a certain level of domain expertise.

4.3 AutoGluon Model

4.3.1 Model Overview

AutoGluon is an AutoML framework designed to efficiently train high-performing models on tabular data [10]. In this implementation, it was used to predict corporate bankruptcy using the same SMOTE-resampled Polish dataset just like the XGBoost model. AutoGluon supports ensemble learning out of the box, combining various base models such as LightGBM, CatBoost, and neural networks into a stacked architecture for improved generalization.

4.3.2 Model Training and Optimization

Training was conducted using AutoGluon's TabularPredictor with the best_quality preset, which prioritizes accuracy through multi-layer ensembling and extensive bagging. No manual hyperparameter tuning was required, as AutoGluon internally evaluated a wide range of model architectures and selected the best-performing ensemble. The final stacked ensemble included LightGBM with extended tuning (LightGBMXT), a Random Forest with entropy-based splitting, and a weighted ensemble model that learned to combine the predictions of its base models. All training was performed on the same SMOTE-resampled dataset used in the XGBoost pipeline to ensure consistency.

4.3.3 Evaluation

The AutoGluon ensemble achieved strong predictive performance on the test set, reaching an overall accuracy of 0.9770. For the minority class representing bankrupt firms, it yielded a precision of 0.7824 and a recall of 0.7225. This indicates

the model can identify a substantial portion of at-risk firms while maintaining a reasonable false positive rate—an essential balance for high-stakes financial risk prediction tasks.

To interpret the model's behavior, permutation importance was used to quantify the influence of individual features on prediction accuracy. The results, shown in Table 5, highlight several financial ratios with notable predictive power. For instance, `predictedrisk` dominates with an importance score of 0.08332 and a narrow 99th percentile range [0.076, 0.0906], suggesting it consistently plays a central role in decision-making across all folds. Among other financial indicators, `rotation_receivables_inventory_turnover_in_days` and `receivables_365_sales` follow closely, each with high importance scores and statistically significant p-values (0.001 and 0.002 respectively), emphasizing their reliability as predictors.

Several other variables—such as `sales_inventory`, `inventory_365_cost_of_products_sold`, and `book_value_of_equity_total_liabilities`—also exhibit moderate importance, though with wider confidence intervals, indicating slightly more variability in their impact. The range and p-values collectively highlight which features are both impactful and stable, which is especially valuable in interpreting AutoML-generated ensembles where model logic is not transparent to the user.

Table 4. Top 10 Features Based on Permutation Importance (AutoGluon)

Feature	Importance	p-value	99 th Percentile Range
<code>current_assets_inventory_short_term_liabilities</code>	0.00024	0.035	[−0.000211, 0.000691]
<code>book_value_of_equity_total_liabilities</code>	0.00032	0.001	[0.000094, 0.000546]
<code>sales_inventory</code>	0.00032	0.070	[−0.000483, 0.001123]
<code>inventory_365_cost_of_products_sold</code>	0.00032	0.039	[−0.000305, 0.000945]
<code>sales_receivables</code>	0.00036	0.035	[−0.000317, 0.001037]
<code>profit_on_sales_total_assets</code>	0.00036	0.061	[−0.000484, 0.001204]
<code>gross_profit_depreciation_sales</code>	0.00036	0.027	[−0.000251, 0.000971]
<code>receivables_365_sales</code>	0.00060	0.002	[0.000096, 0.001104]
<code>rotation_receivables_inventory_turnover_in_days</code>	0.00064	0.001	[0.000295, 0.000985]
<code>predictedrisk</code>	0.08332	≈0.000	[0.076, 0.0906]

4.3.4 Results and Limitations

AutoGluon performed well on the bankruptcy prediction task, producing strong results without requiring manual tuning. Its layered ensembling approach contributed to robust classification and effective handling of class imbalance. However, a notable limitation is its interpretability, as permutation importance provides a high-level understanding of feature relevance but does not offer the localized, case-specific interpretability which is often required in high-stakes financial settings. Additionally, as a fully automated system, it functions largely as a black box introducing a lack of transparency, which may limit its adoption in financial domains where transparency and justification of predictions are crucial.

4.3.5 Key Takeways and Insights

AutoGluon proved to be an effective tool for bankruptcy prediction, delivering strong classification performance with minimal manual intervention. Its ensemble strategy, which combined models such as LightGBMXT and Random Forest through weighted stacking, allowed the system to achieve a high test accuracy of 0.9770 and a recall of 0.7225 on the

minority (bankrupt) class. This indicates a strong ability to detect distressed firms in an imbalanced setting—a critical requirement in financial risk forecasting.

One of AutoGluon’s greatest strengths is its automation of model selection, training, and optimization, making it highly accessible for practitioners. However, while it does offer global interpretability via permutation importance, this technique provides a less-detailed view compared to SHAP-based methods. For instance, although top features like `rotation_receivables_inventory_turnover_in_days` and `receivables_365_sales` emerged as statistically significant predictors, the interpretation remains limited to their average contribution rather than their instance-level impact.

Therefore, while AutoGluon is well-suited for performance-oriented applications, its interpretability limitations may constrain its utility in domains where explainability, auditability, or regulatory justification are essential. In such cases, complementary techniques or more transparent models may be necessary to support decision-making alongside predictive output.

4.4 Model Comparison and Discussion

This section provides a comparative analysis of the XGBoost and AutoGluon models developed for predicting corporate bankruptcy. Both models were trained on the same SMOTE-resampled dataset and evaluated using identical test sets, enabling a fair comparison across core performance metrics and interpretability characteristics.

In terms of classification performance, both models achieved high accuracy on the test set (approximately 97.7% for AutoGluon and 98.0% for XGBoost). However, AutoGluon slightly outperformed XGBoost in terms of recall for the minority (bankrupt) class, detecting 72.25% of bankrupt firms compared to 71.05% in the optimized XGBoost configuration. While this difference is small, it highlights AutoGluon’s advantage in capturing more true positives without the need for manual hyperparameter tuning. Precision scores were also comparable, with AutoGluon achieving 0.7824 versus 0.81 for XGBoost. The metrics for the two models can be seen in Table 6.

Table 5: Metrics Comparison for XGBoost and AutoGluon

Model Implementation	ROC AUC	Accuracy	Precision	Recall	F1 Score
XGBoost	0.972	0.980	0.810	0.710	0.760
AutoGluon	0.859	0.977	0.782	0.722	0.751

One of the major differentiators lies in interpretability. XGBoost, when paired with SHAP, provides rich global and local explanations—enabling a detailed understanding of how each feature contributed to individual predictions. This is especially important in the financial domain, where decision-makers often require transparency to justify predictions. AutoGluon, on the other hand, relied on permutation importance, which offers less detailed insight and does not explain predictions at the instance level. While useful for identifying globally relevant features, permutation importance is limited in contexts that demand interpretability for compliance or auditability.

From a usability perspective, AutoGluon’s AutoML framework significantly reduces the need for expert intervention, automating model selection, training, and ensembling. This makes it highly appealing for rapid prototyping or deployment in resource-constrained environments. XGBoost, in contrast, requires manual tuning and architectural decisions, but offers more control and explainability—an asset in regulated sectors.

In summary, both models are effective for bankruptcy prediction, but serve different priorities. AutoGluon is better suited for performance-focused use cases where automation and speed are critical, while XGBoost offers greater transparency, making it more appropriate for domains requiring accountability and interpretability.

5 CONCLUSION

This study presented a comprehensive exploration of bankruptcy prediction using both traditional and advanced machine learning techniques across two benchmark datasets. Initial experiments using metaheuristic-optimized neural networks on the Taiwanese dataset revealed the limitations of high-accuracy models in imbalanced scenarios. With the application of SMOTE, ADASYN, and GAN resampling techniques, the Random Forest classifier demonstrated significant performance gains, especially in recall.

Building upon these findings, we applied XGBoost and AutoGluon to the Polish dataset for a more interpretable and scalable modeling pipeline. XGBoost provides superior transparency through SHAP values, making it ideal for financial settings requiring explainability. In contrast, AutoGluon achieved competitive results with minimal manual intervention, showcasing the power of AutoML in time-sensitive or resource-constrained environments. The comparative analysis illustrates that the choice of model depends on the deployment context—XGBoost excels in interpretability, while AutoGluon streamlines implementation.

Future work may explore hybrid approaches that combine the interpretability of XGBoost with the automation of AutoGluon, alongside deeper integration of time-series insights and cost-sensitive learning for improved financial forecasting.

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